

K40: T841 P(3) Erratum-Upgrade — c_2 vs C_2 Conflation Sweep

Keeper

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Verdict: ERRATUM, ISOLATED, UPGRADE

T841's registry entry contains a single arithmetic error: $P(3) = g \cdot C_2 = 42$. The correct identity is $P(3) = g \cdot c_2 = 77$, where $c_2 = 11 = \text{rank} \cdot n_C + 1$ is the second Chern class integer (distinct from $C_2 = 6$, the second Casimir / scalar Laplacian gap). The corrected identity uses four BST integers (g , rank , n_C , $+1$ unit) rather than two — this is a strengthening, not a weakening.

Contamination scope: isolated to T841's registry summary. The Q^5 Hilbert polynomial formula in Toy 910 line 255 is correct — it just was not extended to test $P(3)$ in its 8/8 PASS list. Other " $42 = g \cdot C_2$ " appearances in the corpus are in legitimately different contexts (θ -correspondence, eigenvalue product, mass formulas) where $42 = 7 \times 6$ is correct arithmetic for that identity.

Origin

From Toy 2255 (Elie, 2026-05-15, 38/40): - $P(Q^5, 1) = 7 = g$ (T841 claim correct) - $P(Q^5, 2) = 27 = N \cdot c^3$ (T841 claim correct, K38 load-bearing leg confirmed) - $P(Q^5, 3) = 77 \neq 42$ (T841 claim wrong)

Elie's identification of the correct factorization:

$$P(Q^5, 3) = 77 = g \cdot c_2 = 7 \cdot 11 = 7 \cdot (\text{rank} \cdot n_C + 1)$$

where $c_2 = 11$ is the second Chern class integer of Q^5 , not $C_2 = 6$ (the second Casimir / scalar Laplacian gap on D_{IV}^5).

c_2 vs C_2 — the disambiguation

These are two distinct BST integers that have shared the same notation in older docs:

Symbol	Value	Meaning	Where
c_2	11	Second Chern class of $Q^5 = \text{rank} \cdot n_C + 1 = 2\text{-form}$ Laplacian gap = $\dim K$	T1484, T1788, T1790, T1791, T1794, T1726
C_2	6	Second Casimir invariant = scalar Laplacian gap = $\chi(Q^5)$	T186, all BST core docs

The distinction is already correctly established in newer theorems (T1484 Thirteen Theorem explicitly references " $c_2 \cdot c_3 = 11 \cdot 13 = 143 = N_{\text{max}} + C_2$ "; T1788 YM Ring Uniqueness writes "adjoint gap $c_2/C_2 = 11/6$ "; T1790 Weitzenböck 2-Form Gap states "2-form Laplacian gap on Q^5 is $c_2 = 11$, the second Chern class"; T1791 confirms

“universal identity $c_2 = \dim K$ holds for all quadrics”; T1794 distinguishes “adjoint gap $c_2 = 11$ exceeds scalar gap $C_2 = 6$ ”).

T841 (registered earlier — predates the c_2/C_2 disambiguation) used the symbol C_2 throughout its registry summary line, including a place where it should have written c_2 . The arithmetic mistake was the inevitable consequence.

Sweep results

Confirmed-error files (need fix)

File	Line	Current	Correct
notes/BST_AC_TheoremRegistry.md	89	$P(3)=g \cdot C_2$	$P(3)=g \cdot c_2 = 7 \cdot 11 = 77$

That is the entire confirmed-error scope. ONE entry in ONE file.

Verified-correct files (no fix needed)

File	Use of “ $42 = g \cdot C_2$ ”	Why correct
BST_22_UniquenessConditions.md:229	Correctly uses product $d_1 \cdot \lambda_1 = g \cdot (n+1) = 42$	This is the eigenvalue-multiplicity $\times$ eigenvalue product on Q^5 (NOT the Hilbert polynomial value). Substituting $n=5$ gives $7 \times 6 = 42$. Correct identity for THIS quantity.
BST_22_UniquenessConditions.md:229	Correctly uses $42 = 42$ only at $n=5$	Same product, level-rank duality framing. Correct arithmetic for the $SO(7)_2 \times Sp(6)_2$ central charge product.
notes/.running/g2ace_ggf_product_theorem.md:36	Correctly uses $42 = 7 \cdot 6$ for matter mode count.	Number-theoretic “matter mode” decomposition. Arithmetic $42 = 7 \times 6$ is correct.
play/toy_1318_artifact_pickets Enumeration 42:237	Correctly uses $42 = 42$	θ -correspondence dimension of $Sp(6)$ representation. Standard. Not Hilbert poly.
play/toy_1857_darDMatGenValZackmShadow.py:426	Correctly uses $42 = \pi^5 \cdot m_e$	Mass formula using $g \cdot C_2$ as a numerical coefficient. Different physical context.
MESSAGES_2026-05-03:437	“Hilbert-to-eigenvalue ratio $P(1)/\lambda_1 = g/C_2$ ”	This is $P(1)=g$ over $\lambda_1=C_2$. $P(1) = 7$ (Q^5 Hilbert) over $C_2 = 6$ (Casimir). Both readings consistent with Toy 2255.

Ambiguous (flagged for Lyra clarification)

BST_22_UniquenessConditions.md:61 includes the phrase: “*The value 42 is the dimension of the θ -correspondence representation of $Sp(6)$ and the Hilbert series evaluation $P(1)$...*”

The “Hilbert series evaluation $P(1)$ ” sub-clause is ambiguous: if it refers to the Q^5 Hilbert polynomial, it is wrong ($P(Q^5, 1) = 7$, not 42, per Toy 2255). If it refers to a different Hilbert series (e.g., the $Sp(6)$ θ -correspondence Hilbert series, which IS dimension 42 at the relevant degree), it is correct.

Recommendation: Lyra clarify the sub-clause to specify *which* Hilbert series, OR strike it as conflated wording. Severity: MINOR.

Why this is an UPGRADE, not just a fix

T841 claimed: $P(3) = g \cdot C_2 = 7 \cdot 6 = 42$, using two BST integers.

Corrected identity: $P(3) = g \cdot c_2 = g \cdot (\text{rank} \cdot n_C + 1) = 7 \cdot (2 \cdot 5 + 1) = 7 \cdot 11 = 77$, using four BST integers plus a unit ($g, \text{rank}, n_C, +1$).

The corrected form binds more of the BST integer cascade into a single Hilbert polynomial value. It is also consistent with the structural pattern Lyra identified post-Toy 2255:

- $c_2 = \text{rank} \cdot n_C + 1 = 11$ (second Chern integer, the “+1” is genus correction at low degree)
- $N_{\text{max}} = N_{c^3} \cdot n_C + \text{rank} = 137$ (spectral cap, the “+rank” is genus correction at the maximal K-type level)

Same Hilbert-polynomial structure: coefficient $\times n_C$ + genus correction. This is the basis for Lyra’s reframed A2 search — instead of looking for a single +rank derivation, look for the family of shifts $c_k = [\text{coefficient}_k] \cdot n_C + [\text{genus_correction}_k]$ that the Bergman kernel exponent on D_{IV}^5 generates, and read off rank at level $k = N_{c^3}$.

The erratum is therefore not just an arithmetic fix — it surfaces a STRUCTURAL PATTERN that Lyra now has as a probe for A2.

Actions

#	Action	Owner	Status
K40-1	Fix T841 registry entry line 893: change $P(3)=g \cdot C_2$ to $P(3)=g \cdot c_2 (= 77)$	Keeper	THIS SESSION
K40-2	Annotate T841 entry with K40 erratum reference	Keeper	THIS SESSION
K40-3	Flag <code>BST_22_Uniqueness_Conditions.md:61</code> sub-clause for Lyra clarification	Lyra (when next on docs)	QUEUED
K40-4	Tag Hilbert polynomial entries in <code>data/bst_geometric_invariants.json</code> with K40 reference (Hilbert_Q5_P3 should note T841 erratum)	Grace	DONE (per RUN_LIST T1.8)
K40-5	Add c_2 vs C_2 disambiguation note to onboarding docs (<code>CLAUDE.md</code> or <code>bst_seed.md</code>)	Keeper or Grace	OPTIONAL

Implication for K38

K40 does NOT downgrade K38. The chain N_{c^3} leg rests on $P(Q^5, 2) = 27$, verified by Toy 2255 independently of any T841 claim. K38 retains its A1-PASS upgrade.

K40 strengthens K38 indirectly: the $c_2 = \text{rank} \cdot n_C + 1$ structural pattern is exactly the kind of “genus correction at a forced level” that Lyra needs to derive for the +rank step of K38 (A2). If A2 succeeds along that line, the K40 erratum will have surfaced the very mechanism that closes K38 external D-tier.

Bottom line

T841 had a single arithmetic error inherited from pre-disambiguation c_2/C_2 notation. The correct identity is stronger than the original claim. Contamination is isolated to one registry entry. The fix produces a structural

pattern that points the way to closing A2.

Erratum filed. K38 stands. Lyra's A2 has a new probe.

— Keeper, K40, 2026-05-15